

Theory Only

Prabinchy (Python)
NAST

Baye's Theorem:

* statement:

Let $E_1, E_2, E_3, \dots, E_n$ be the mutually disjoint of event of a sample space S with $P(E_i) \neq 0, i = 1, 2, 3, \dots, n$ then for any arbitrary event A which is a subset of S such that $P(A) > 0$, we write

$$P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{\sum_{i=1}^n P(E_i) \cdot P(A/E_i)}$$

Proof:

Since A is a subset of S and

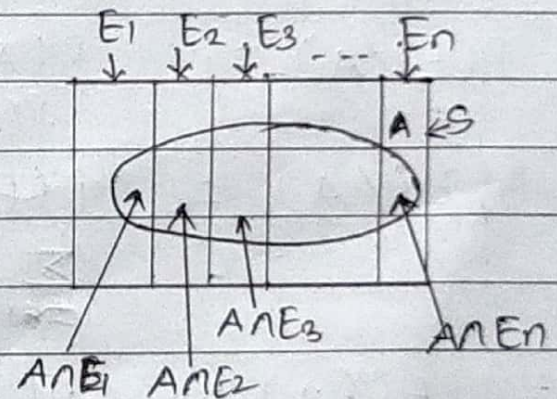
~~so we write~~

$$S = E_1 \cup E_2 \cup E_3 \dots \cup E_n, \text{ so}$$

we write,

$$A = A \cap S$$

$$\Rightarrow A = A \cap [E_1 \cup E_2 \cup E_3 \dots \cup E_n]$$



$$\Rightarrow A = (A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \cup \dots \cup (A \cap E_n)$$

➔ Taking probability on both sides, we get;

$$P(A) = P[(A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \cup \dots \cup (A \cap E_n)]$$

$$\Rightarrow P(A) = P(A \cap E_1) \cup P(A \cap E_2) \cup P(A \cap E_3) \cup \dots \cup P(A \cap E_n)$$

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Then for any particular event, $(A \cap E_i)$,

$$P(A \cap E_i) = P(A) \cdot P(E_i/A)$$

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This is the required

Hence Baye's theorem is proved.

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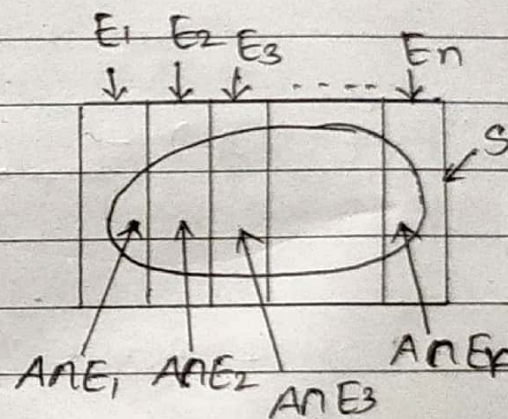
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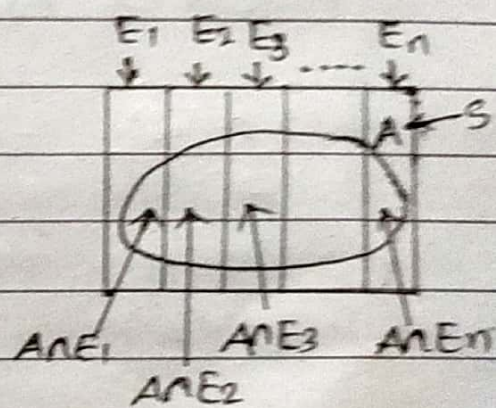
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Criteria of good estimator:

Prabin Chaudhary, NAST

→ A good estimator is one which is very close to the true value of population parameter as much as possible.

→ A good estimator has following criterias:

1. Unbiasedness:

→ A statistic $\hat{\theta}$ is said to be unbiased estimator of a population parameter θ , if and only if the mean of the sampling distribution is equal to θ whatever the value of θ .

2. Consistency:

→ A statistic $\hat{\theta}$ is said to be consistent estimator of a population parameter θ , if as the sample size increases, the value of the sample statistic becomes very close to the value of population parameter.

3. Efficiency:

→ A statistic $\hat{\theta}$ is said to be efficient estimator of a population parameter θ , if the variance of the estimator $\hat{\theta}$ is least with comparison to the variance of all the other estimator of θ .

4. Sufficiency:

→ A statistic $\hat{\theta}$ is said to be sufficient estimator of a population parameter θ if it contains all the information in the sample about the parameter.

→ A sufficient estimator will also be the most efficient and the most consistent estimator.

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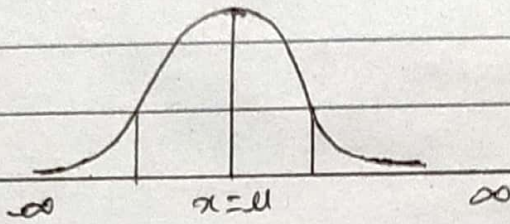
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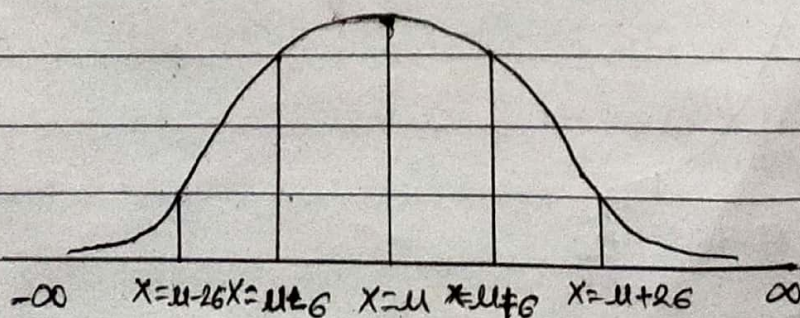
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Characteristics of Normal distribution: Prabin chand hary, NAST

- i> A normal distribution is a continuous probability distribution and has two parameters μ and σ .
- ii> In normal distribution, the curve is bell shaped and symmetrical about the line $x = \mu$.

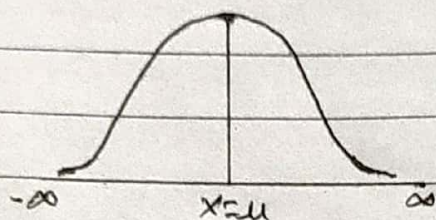


- iii> The normal distribution is symmetrical as mean = median = mode = μ .
- iv> In normal distribution, mean = μ , standard deviation = σ , skewness = 0 and kurtosis = 3.
- v> In normal distribution, the quartiles are equidistant from mean (μ), $Q_1 = \mu - 0.6745\sigma$ and $Q_3 = \mu + 0.6745\sigma$.
- vi> In normal distribution, the maximum probability of the curve occurs at $x = \mu$ and maximum $f(x) = \frac{1}{\sigma\sqrt{2\pi}}$.
- vii> In normal distribution, the point of inflection of the curve occurs at $x = \mu \pm \sigma$.
- viii> In normal distribution, the x-axis is asymptote to the curve.
- ix> Area property:

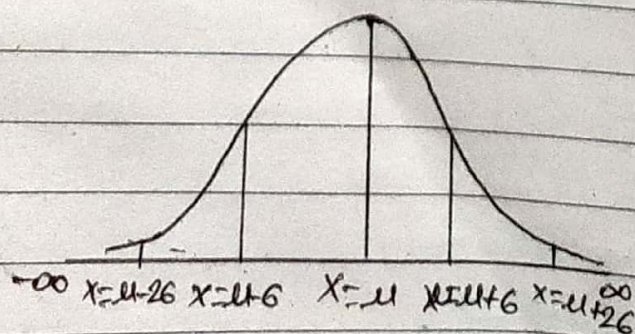


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Error in Hypothesis Testing:

- The decision to accept or reject a null hypothesis H_0 after inspection of samples of which are wrong or right decisions.
- There are four possible decisions which are
 - i) Reject H_0 when H_0 is true
 - ii) Accept H_0 when H_0 is true
 - iii) Reject H_0 when H_0 is false
 - iv) Accept H_0 when H_0 is false
- The decisions (ii) and (iii) are correct decisions and (i) and (iv) are wrong decisions.
- These decisions can be expressed in the following dichotomous table.

	Accept H_0	Reject H_0
H_0 true	correct	Wrong (I)
H_0 false	wrong (II)	correct

- The error of rejecting H_0 when H_0 is true is called Type I error and it is denoted by α .
- The error of accepting H_0 when H_0 is false is called Type II error and it is denoted by β .

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→ The decisions (i) and (ii) are wrong decisions while

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